

Knowing but Not Saying: Probe-Saturation Gating and Engagement Schedules for the Knowing–Using Gap in Fine-Tuned Language Models

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Abstract

When a language model is fine-tuned on new factual content, two capabilities diverge: the model quickly becomes able to recall an answer when the question is posed in the training format (knowing), while its ability to produce the answer under distribution shift lags far behind (using). We introduce MLPR, a lightweight monitoring-and-regularization layer that makes this knowing–using gap measurable and actionable during training: a linear probe at a mid layer l^* reads entity identities from hidden states, and a saturation gate ramps in a probe-alignment loss $\lambda(t)$ once a monitored signal crosses a threshold τ_0 . We study Qwen2.5-0.5B and Qwen3-1.7B under an engagement-schedule grid of ten fully instrumented runs on a synthetic entity-question corpus, comparing two gate signals (output-likelihood recall vs. representation-space probe decodability) and three engagement schedules (sustained, decayed, output-triggered). Three findings emerge. First, the gate signal decides when engagement is possible: the probe signal opens the gate at epoch 3–6 at both scales, 11–16 epochs before the output signal crosses its own threshold in the replicates where it ever does, and in one output-gated replicate it never crosses within budget. Second, the engagement schedule decides whether memorization completes: late output-triggered engagement reaches SFT recall parity (0.926 vs. 0.934; paired McNemar $p=0.83$) and completes a previously budget-censored regularized phase with 8 stable epochs at full probe weight, whereas early sustained injection is high-variance (0.109–0.805) and transient injection with decay to zero collapses recall to 0.012 while the representation still decodes 94.8% of entities—the sharpest knowing–saying dissociation we observe. Third, free-generation and generalization exact match stay at $\leq 0.4\%$ in every arm at both scales even as teacher-forced recall reaches 0.93: the saying gap is scale-invariant, and all parameter updates concentrate in upper-layer MLP projections. Code, telemetry, and trained adapters are released.

1 Introduction

Fine-tuning is the standard route for injecting new knowledge into pretrained language models, whether through supervised instruction tuning, domain adaptation, or parameter-efficient updates (Hu et al., 2022). A growing body of evidence shows that this process memorizes far more readily than it generalizes: models fit training-format facts quickly (Tirumala et al., 2022; Carlini et al., 2023), while behavioral control of the acquired knowledge—answering paraphrases, chaining facts, or refusing when unsure—arrives late or not at all (Kandpal et al., 2023; Allen-Zhu & Li, 2023). For practitioners this gap is expensive: a model that knows an entity (its answer is recoverable from its internal states) but cannot say it under mild distribution shift will silently fail downstream, and standard observables cannot distinguish the two regimes.

We propose to make this distinction measurable during training, and to act on it. Our approach, MLPR (Mid-Layer Probe Regularization), wraps a parameter-efficient fine-tuning run with three coupled instruments. First, a small linear probe p_ϕ attached to a mid layer l^* reads entity identities from the backbone’s hidden states; because the probe trains on

detached states during a warm-up phase, it is a pure observer that cannot steer the backbone. Second, a teacher-forced likelihood metric estimates recall saturation $A_{\text{mem}}^{\text{lf}}$ each epoch by scoring the answer tokens given the training-format question—a “knowing” signal that, unlike free-generation exact match, does not conflate knowledge with answer extraction. Third, a saturation gate compares a monitored signal against a threshold τ_0 and, once crossed, ramps in a probe-alignment loss with weight $\lambda(t)$ so that the backbone is regularized toward keeping acquired identities linearly decodable at the mid layer. An eval-aware adaptive controller completes the loop by damping the learning rate when held-out loss degrades, subject to an explicit floor that keeps memorization dynamics alive.

We instantiate MLPR on Qwen2.5-0.5B-Instruct and Qwen3-1.7B (Qwen Team, 2024; Yang et al., 2025), training a grid of ten fully instrumented runs (24 epochs; two 16-epoch censored replicates) on a synthetic entity-question corpus with a 1,600-entity vocabulary (per-step losses, per-epoch gate evaluations, per-sample final evaluations, adapters and probe checkpoints). The grid crosses two gate signals—output-likelihood recall vs. representation-space probe decodability—with three engagement schedules (sustained, decayed-to-zero, output-triggered), completing the engagement-schedule arm that was budget-censored in our earlier large-scale study. The resulting dataset of training dynamics yields findings we summarize here and develop in Section 5:

- The gate signal decides when engagement is possible (Section 5.1). The probe’s representation decodability crosses its threshold at epoch 3 (0.5B) and epochs 4–6 (1.7B), while the output-likelihood signal crosses $\tau_0=0.8$ only at epoch 17–19—and in one output-gated replicate never crosses at all. Gating on the representation is scale-aware by construction; gating on the output is reachable at scale but unstable in crossing time.
- The engagement schedule decides whether memorization completes (Section 5.2). Under identical data, seed, and optimizer, final recall spans 0.012–0.934 at 1.7B. Late output-triggered engagement reaches SFT parity (0.926 vs. 0.934, McNemar $p=0.83$) and completes the regularized phase—8 stable epochs at full probe weight—that our earlier large-scale run left budget-censored. Early sustained injection is high-variance (0.109, 0.805); transient injection that decays to zero collapses recall to 0.012 while the probe still decodes 94.8% of entities.
- Knowing and saying dissociate at both scales (Section 5.3). Teacher-forced recall reaches 0.91–0.93 and token-level recall 0.99, yet free-generation EM stays ≤ 0.012 and generalization EM ≤ 0.004 in every arm: the saying gap is invariant across a $3.4\times$ parameter range and across all schedules.
- Probes detect memorization early and cheaply (Section 5.4). The probe crosses 0.5 decodability between epochs 4 and 7 everywhere, leading output recall by 8–18 epochs (and leading arms whose output never crosses at all), with loss two orders of magnitude below chance.
- Updates concentrate in upper-layer MLPs (Section 5.5). $\Delta W = \frac{\alpha}{r}BA$ per LoRA-adapted projection puts 65–71% of update mass in MLP projections, key/value attention nearly untouched, consistent with factual storage accounts (Meng et al., 2022; Geva et al., 2021).

Our contributions are: (i) a training-time instrumentation stack—probe, likelihood-based recall metric, saturation gate with selectable signal, and engagement schedules—that turns the knowing–using gap into an observable, thresholdable quantity; (ii) a completed engagement-schedule grid at two scales with paired McNemar analysis, providing the first fully exercised regularized phase (8 epochs at full probe weight) and its outcome parity with SFT; (iii) a dissociation analysis separating representation decodability, teacher-forced recall, free-generation EM, and chaining/intersection transfer, including a representation-without-output regime induced by transient engagement; and (iv) engineering lessons from instrumenting real runs, including a documented failure mode and repair for eval-aware learning-rate controllers (Appendix D). All telemetry, code, and trained adapters are released.

2 Related Work

Probing internal representations. Linear probes are a standard instrument for asking what models encode (Alain & Bengio, 2016; Belinkov, 2022); control tasks guard against probes that exploit surface statistics (Hewitt & Liang, 2019). Our probe differs in role: trained alongside the run on detached states, its readings drive a control decision—activating a regularization pathway—turning it from a diagnostic into a controller input. Richer readouts (Bricken et al., 2023) could substitute in future work.

Knowledge storage and memorization dynamics. Mechanistic and editing studies localize factual knowledge in mid-to-upper MLP blocks that behave as key-value memories (Meng et al., 2022; 2023; Geva et al., 2021; Dai et al., 2022; Mitchell et al., 2022); our anatomy analysis is consistent with this picture from the direction of training dynamics. Synthetic entity corpora with known ground truth have been used to study capacity and retrieval (Allen-Zhu & Li, 2023); our generator follows this tradition but exposes per-epoch recall dynamics. Memorization of training data is well documented (Carlini et al., 2023; Zhang et al., 2017), large models memorize more before they overfit in loss terms (Tirumala et al., 2022), and knowledge acquisition is strongly scale-dependent (Kaplan et al., 2020; Kandpal et al., 2023); our grid gives a fine-grained per-epoch view in the fine-tuning regime. The dissociation we measure extends paraphrase-consistency findings (Elazar et al., 2021; Mallen et al., 2023) into training dynamics, and recalls the competence-expression decoupling of grokking (Power et al., 2022), here driven by data format rather than weight decay.

Adaptive training control. Adaptive learning-rate schedules respond to loss plateaus or gradient statistics; our controller adds a held-out-loss criterion with a twist specific to the memorization regime: because rising held-out loss is the very signal the gate waits for, damping must be bounded lest the controller extinguish the phenomenon it monitors. We are not aware of prior work analyzing this controller-phenomenon interaction; we document a negative result (an unbounded decay spiral) and its repair.

3 Method: Mid-Layer Probe Regularization

MLPR augments parameter-efficient fine-tuning with an observation pathway (a mid-layer probe and an online recall metric) and a control pathway (a saturation gate and an eval-aware controller). Figure 1 shows the data flow; Algorithm 1 gives the training loop.

3.1 Setup and objective

Let f_θ be a pretrained decoder-only LM, and let LoRA (Hu et al., 2022) add low-rank updates $\Delta W = \frac{\alpha}{r}BA$ to selected weight matrices, with A, B trainable and θ frozen. The supervised objective on the memorization corpus $\mathcal{D}_{\text{mem}}$ of question-answer pairs is the token-level cross-entropy

$$\mathcal{L}_{\text{ce}} = -\frac{1}{|B|} \sum_{(x,y) \in B} \log p_\theta(y | x), \quad (1)$$

where x is the training-format question and y the answer with Answer: suffix. At each optimizer step the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda(t) \mathcal{L}_{\text{probe}}(p_\phi, f_\theta), \quad (2)$$

where $\lambda(t) \in [0, \lambda_0]$ is the gate weight (Section 3.3) and $\mathcal{L}_{\text{probe}}$ is a cross-entropy between the probe’s entity prediction and the ground-truth entity id of the sample in the batch, computed on the mid-layer hidden state h_{l^*} with gradients flowing into the backbone once $\lambda > 0$ (before activation the probe receives no gradient path into θ , cf. Eq. 2 with $\lambda=0$).

3.2 Probe warm-up and the likelihood recall metric

The probe $p_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{V_e}$ is a single linear layer over the entity vocabulary V_e (here $|V_e|=1600$), trained from the first step on detached states at $l^* \approx L/2$ with its own learning

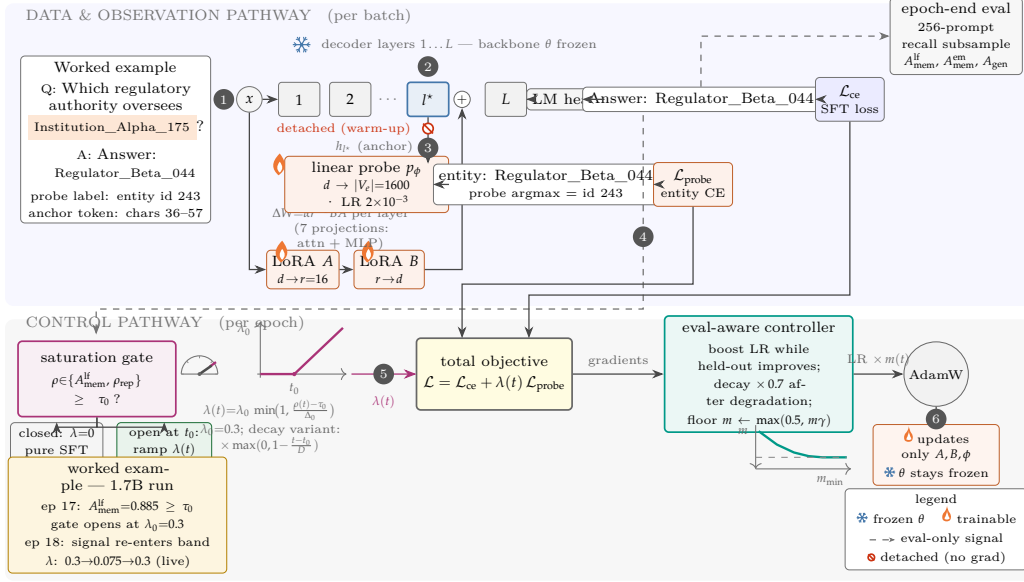


Figure 1: MLPR methodology with a worked example. Top — data & observation pathway (per batch): a memorization sample from $\mathcal{D}_{\text{mem}}$ (left card; anchor token highlighted) is processed by the frozen backbone (snowflake: θ receives no gradient) augmented with trainable low-rank adapters (flame: A, B , $r=16$, applied to all seven attention and MLP projections, merged at $\oplus$). The mid-layer state h_{l^*} at the anchor token is read by a linear probe p_ϕ (flame) that predicts the entity identity; during warm-up the probe observes a detached copy of the state ($\odot$, no gradient into θ). Bottom — control pathway (per epoch): the saturation gate (gauge) compares a monitored signal ρ —teacher- forced recall $A_{\text{mem}}^{\text{lf}}$ or the probe’s representation decodability ρ_{rep} —against τ_0 ; once it crosses, $\lambda(t)$ ramps the probe-alignment loss into the total objective (ramp inset), with an optional linear decay to zero (λ -decay schedule). Numbered badges trace the example: (1) sample enters; (2) frozen+LoRA forward; (3) detached mid-layer probe read; (4) epoch- end recall evaluation (256-prompt subsample); (5) gated loss weight; (6) optimizer updates trainable parameters only. The amber card reports the realized 1.7B output-gated run: the gate opened at epoch 17 ($A_{\text{mem}}^{\text{lf}}=0.885 \geq \tau_0$), and the live per-epoch re-evaluation is visible when the signal re-enters the band at epoch 18 and λ drops $0.3 \rightarrow 0.075$ before re-ramping.

rate (2×10^{-3} vs. 2×10^{-4} for the backbone), so that it converges quickly enough to be informative early. Because the probe gradient is detached during warm-up, it is a pure observer; only after gate activation does $\mathcal{L}_{\text{probe}}$ back-propagate into θ , at which point it encourages the backbone to keep acquired entity identities linearly decodable at l^* .

We measure recall with a teacher-forced likelihood probe: for each memorization prompt (answer suffix stripped), the answer tokens are scored in a single forward pass and the greedy per-sample match is

$$\text{lf}(x, y) = \mathbb{1}[\arg \max_w p_\theta(w | x) = y_1], \quad (3)$$

with a token-level variant that credits partial answers. The epoch-level recall is $A_{\text{mem}}^{\text{lf}} = \mathbb{E}_{\mathcal{S}}[\text{lf}]$ on a fixed subsample $\mathcal{S}$ of $\mathcal{D}_{\text{mem}}$ (256 prompts). We additionally log the free-generation exact match $A_{\text{mem}}^{\text{em}}$, substring containment, chaining/intersection generalization accuracy A_{gen} on $\mathcal{D}_{\text{gen}}$, and the probe’s own entity-decodability on memorized states, giving a multi-channel view of the same underlying state change.

Algorithm 1: MLPR training loop (one epoch)

Input: corpora; LoRA backbone f_θ ; probe p_ϕ ; gate (4); controller with floor (6)

- 1 probe warm-up on detached h_{l^*} ;
- 2 foreach batch $B \subset \mathcal{D}_{\text{mem}}$ do
- 3 $\mathcal{L} \leftarrow \mathcal{L}_{\text{ce}} + \lambda(t) \mathcal{L}_{\text{probe}}$; update $\{A, B, \phi\}$;
- 4 $A_{\text{mem}}^{\text{lf}} \leftarrow (3)$ on subsample; evaluate $A_{\text{mem}}^{\text{em}}, A_{\text{gen}}$;
- 5 open gate if $\rho \geq \tau_0$; clamp controller $m \geq m_{\text{min}}$; log;

151 3.3 Saturation gate, $\lambda(t)$, and engagement schedules

152 Let t index epochs. The gate monitors a signal $\rho(t)$ and applies the same smooth-threshold
 153 rule at every evaluation,

$$\lambda(t) = \lambda_0 \cdot \min\left(1, \max\left(0, \frac{\rho(t) - \tau_0}{\Delta_0}\right)\right), \quad (4)$$

154 so λ is 0 below τ_0 , ramps linearly across the band $[\tau_0, \tau_0 + \Delta_0]$, and holds at λ_0 above
 155 it; the per-epoch re-evaluation makes engagement follow the signal in both directions (a
 156 dip back into the band lowers λ). Two signals ρ are studied. The output-likelihood gate
 157 uses the proposal-faithful $\rho = A_{\text{mem}}^{\text{lf}}$ with $\tau_0=0.8$, $\Delta_0=0.05$: engagement is deferred until
 158 the model can actually recall the facts, which makes onset data-dependent and, we find,
 159 stochastic across replicates. The probe-saturation gate uses the probe’s entity-decodability
 160 on memorized mid-layer states, $\rho = \rho_{\text{rep}}$ with $\tau_0=0.35$, $\Delta_0=0.15$: the signal is scale-aware by
 161 construction (a fresh linear head reflects representation organization, not LM-head surface
 162 learning), cheap (one forward pass), and the causally correct variable for MLPR—“the
 163 representation knows” is exactly what the probe gradient must preserve. Finally, the λ -
 164 decay schedule modulates any gate by a linear envelope after onset t_0 ,

$$\lambda(t) \leftarrow \lambda(t) \cdot \max\left(0, 1 - \frac{t - t_0}{D}\right), \quad D = 8, \quad (5)$$

165 giving a transient engagement window that closes entirely; sustained engagement is Eq. 4
 166 alone. Together this yields the engagement-schedule grid of Section 4: sustained vs. decayed
 167 engagement under each signal, with $\lambda_0=0.3$ throughout.

168 3.4 Eval-aware adaptive control with a damping floor

169 Long memorization runs exhibit a characteristic trajectory: training loss plateaus after the
 170 format is learned, and held-out loss—bottoming early—then rises monotonically as capacity
 171 is redirected to training entities (Tirumala et al., 2022). An adaptive controller that treats
 172 the rise as divergence may decay the learning rate without bound; the controller we ship
 173 clamps the cumulative LR multiplier applied by eval-driven decays,

$$m(t) \leftarrow \max(m_{\text{min}}, m(t^-) \cdot \gamma), \quad (6)$$

174 with $m_{\text{min}}=0.5$ and $\gamma=0.7$. The floor is the load-bearing fix: held-out divergence is the
 175 phenomenon under study, not a fault, and unbounded damping freezes it (Appendix D
 176 documents the spiral we observed without the floor and its validation sequence). Every
 177 decision is logged per epoch, and all controller actions are failure-isolated. In the production
 178 grid of this paper the LR schedule is a fixed cosine, so the engagement schedule is the only
 179 active intervention; the controller ships with the stack and the floor was exercised and held
 180 in our earlier 50-epoch-class production runs.

181 4 Experimental Setup

182 Corpus. We use a synthetic entity-question corpus in the spirit of controlled knowledge-store
 183 studies (Allen-Zhu & Li, 2023): a vocabulary of 1,600 invented entities with deterministic,
 184 single-token-friendly answers. The memorization split $\mathcal{D}_{\text{mem}}$ contains 1,200 question-answer
 185 pairs in a fixed training template with an Answer: suffix (stripped for all evaluations);

	gate signal	onset	λ_f	$A_{\text{mem}}^{\text{lf}}$	A_{gen}
0.5B SFT	—	—	0.0	0.914	0.012
0.5B MLPR	probe	3	0.3	0.922	0.000
1.7B SFT	—	—	0.0	0.934	0.000
1.7B out-gate (rep. 1)	$A_{\text{mem}}^{\text{lf}}$	never	0.0	0.227	0.000
1.7B out-gate (rep. 2)	$A_{\text{mem}}^{\text{lf}}$	17	0.3	0.926	0.000
1.7B probe-gate (rep. 1)	probe	5	0.3	0.805	0.004
1.7B probe-gate (rep. 2)	probe	6	0.3	0.109	0.000
1.7B λ -decay	probe	5 (decays)	0.0	0.012	0.000

Table 1: Run endpoints (final $n=256$ evaluations). “Onset” is the first epoch with $\lambda > 0$; out-gate rep.1 never crossed $\tau_0=0.8$ (max signal 0.198). All arms reach at most 0.4% generalization EM while teacher-forced recall spans 0.012–0.934: the saying gap is scale-invariant, and the engagement schedule—not the scale—decides whether the recall phase completes.

the generalization split $\mathcal{D}_{\text{gen}}$ (450 prompts) re-asks the trained facts through chaining and intersection templates never seen in training, so exact match on it measures use rather than recall. All entities are synthetic strings, so pre-training leakage is excluded by construction (Appendix B).

Models and training. We train Qwen2.5-0.5B-Instruct and Qwen3-1.7B (Qwen Team, 2024; Yang et al., 2025) with LoRA ($r=16$, $\alpha=32$, dropout 0.05) on all seven attention and MLP projections (8.8M / 17.4M trainable parameters), probe layers $l^*=12/24$ and $14/28$, probe head $d \rightarrow 1600$, backbone LR 2×10^{-4} , probe LR 2×10^{-3} (cosine, 5% warmup), effective batch 16 (75 steps/epoch), 24 epochs, seed 42, bf16, one A100-40GB GPU per run on Modal. The probe optimizer group is gradient-isolated from the backbone clip. Every run emits per-step losses, per-epoch gate evaluations, final per-sample evaluation records, and adapter/probe checkpoints to a persistent volume and to Weights & Biases.

Arms. At 0.5B we run the two-condition comparison: pure SFT ($\lambda \equiv 0$) and probe-gated MLPR. At 1.7B we complete the engagement-schedule grid (the arm that was budget-censored in our earlier study): SFT; output-gated MLPR (gate signal $\rho = A_{\text{mem}}^{\text{lf}}$, $\tau_0=0.8$, $\Delta_0=0.05$; two replicates); probe-gated MLPR (signal ρ_{rep} , $\tau_0=0.35$, $\Delta_0=0.15$; two replicates plus one 16-epoch censored replicate); and a λ -decay arm whose probe weight decays linearly to zero over $D=8$ epochs after onset. Two further 16-epoch censored runs replicate the early-time trajectories (Appendix C).

Metrics. Each epoch we log teacher-forced recall $A_{\text{mem}}^{\text{lf}}$ (3), token-level recall, free-generation EM $A_{\text{mem}}^{\text{em}}$, chaining/intersection EM A_{gen} , and the probe’s entity-decodability ρ_{rep} . Each run ends with a final $n=256$ per-channel evaluation whose prompt-aligned per-sample records feed a paired McNemar test (McNemar, 1947) between arms (Table 1 lists the endpoints of the seven completed 24-epoch runs).

5 Results

5.1 When the gate can open is decided by the gate signal

Figure 2 contrasts the two gate signals. (a) At 0.5B the probe-saturation gate opens at epoch 3: the probe already decodes 40.6% of memorized entities from the mid layer (crossing $\tau_0=0.35$) while output recall is still 0.016; λ reaches $\lambda_0=0.3$ at epoch 4 and holds for 21 epochs. The same arm’s output signal does eventually cross $\tau_0=0.8$ —but only at epoch 19, sixteen epochs later, and in our earlier 50-epoch runs at $4 \times$ lower backbone LR the 0.5B output signal never approached the threshold at all (max 0.39). (b) At 1.7B the output signal is reachable, but its crossing time is unstable: one replicate first exceeds τ_0 at epoch 17 ($A_{\text{mem}}^{\text{lf}}=0.885$); a replicate of the identical configuration never exceeds 0.198. The crossing replicate also demonstrates the gate’s live re-evaluation: at epoch 18 the signal dips

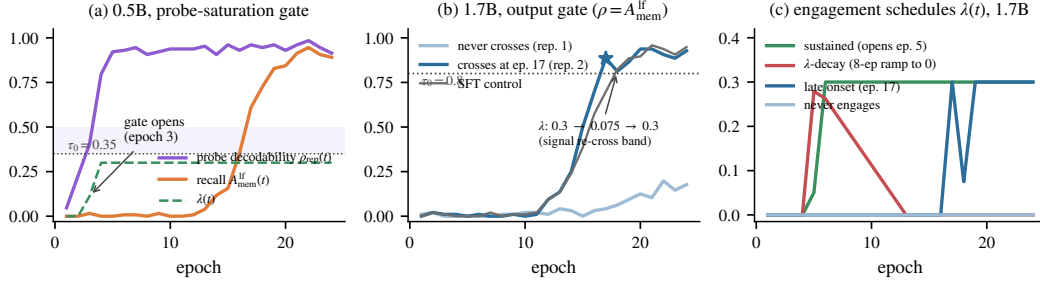


Figure 2: Gate signals and engagement schedules, all curves from run telemetry. (a) 0.5B probe-gated arm: representation decodability ρ_{rep} crosses $\tau_0=0.35$ at epoch 3 while output recall is still ≈ 0 ; $\lambda(t)$ engages and holds. (b) 1.7B output-gated arms ($\rho = A_{\text{mem}}^{\text{lf}}$, $\tau_0=0.8$): one replicate crosses at epoch 17 (star) with the live re-ramp transient $\lambda: 0.3 \rightarrow 0.075 \rightarrow 0.3$ (signal re-cross band); the other never crosses within 24 epochs. (c) The four 1.7B engagement schedules.

back into the $[\tau_0, \tau_0 + \Delta_0)$ band and λ drops $0.3 \rightarrow 0.075$ before re-ramping—the mechanism tracks the signal, not a schedule.

5.2 The engagement schedule decides whether memorization completes

Figure 3 shows the payoff of the 1.7B grid: five arms, identical data, seed, and optimizer, differing only in how the probe loss engages. Three regimes emerge. (i) Late, output-triggered engagement reaches SFT parity. The arm that opened at epoch 17 finished at $A_{\text{mem}}^{\text{lf}}=0.926$ vs. the SFT control’s 0.934 (McNemar $b=12$, $c=10$, $p=0.83$), and exercised the regularized phase for 8 full epochs at $\lambda_0=0.3$: recall rose from 0.698 at activation to 0.926 at the final evaluation with no instability. The phase our earlier large-scale run truncated after 3 epochs is here completed, stable, and harmless to recall. (ii) Early sustained probe injection is high-variance. The two sustained replicates opened the gate one epoch apart (5 vs. 6) yet finished 0.805 and 0.109, both below SFT (rep.1: $b=40$, $c=7$, $p=1.1 \times 10^{-6}$; rep.2: $b=211$, $c=0$, $p=6 \times 10^{-64}$): early probe gradients do not accelerate output-level memorization at 1.7B. (iii) Transient engagement is the worst outcome. The λ -decay arm engaged for 8 epochs, then lost the probe loss entirely; its recall finished at 0.012—an order of magnitude below every other arm—while its representation decodability reached 0.948 (Section 5.3). Finally, the never-engaged replicate doubles as an SFT replication (its probe is detached, $\lambda \equiv 0$): its takeoff lagged the control by about seven epochs and it was still mid-transition (0.227, rising) at epoch 24. Memorization onset at this scale is therefore stochastic across runs—precisely why engagement should be gated on a monitored signal, and why the signal must be one that actually fires.

5.3 Knowing is not saying, at either scale

Figure 4a reports the endpoint channels for all arms. Teacher-forced recall reaches 0.91–0.93 at both scales (and 0.81 for the sustained replicate), token-level recall 0.99, yet free-generation EM is ≤ 0.012 in every arm and generalization EM ≤ 0.004 . The saying gap is invariant across a $3.4\times$ parameter range and across all engagement schedules; regularizing the mid layer does not, by itself, close it. Figure 4b plots the sharpest dissociation in the grid, the λ -decay arm: its probe decodability climbs to 0.948 (the representation organizes the entities essentially perfectly) while output recall never leaves 0.012. Knowing—in the strong, linearly decodable sense—is demonstrably present without saying; the two capabilities are separable training targets, not two ends of one curve.

5.4 Dynamics and the probe as early-warning monitor

Figure 5a shows the shared dynamics. CE plateaus at ≈ 1.4 (0.5B) and ≈ 2.7 (1.7B) once the template is learned, then drops late (epochs 12–16 at 1.7B) to 0.04–0.09 in the arms

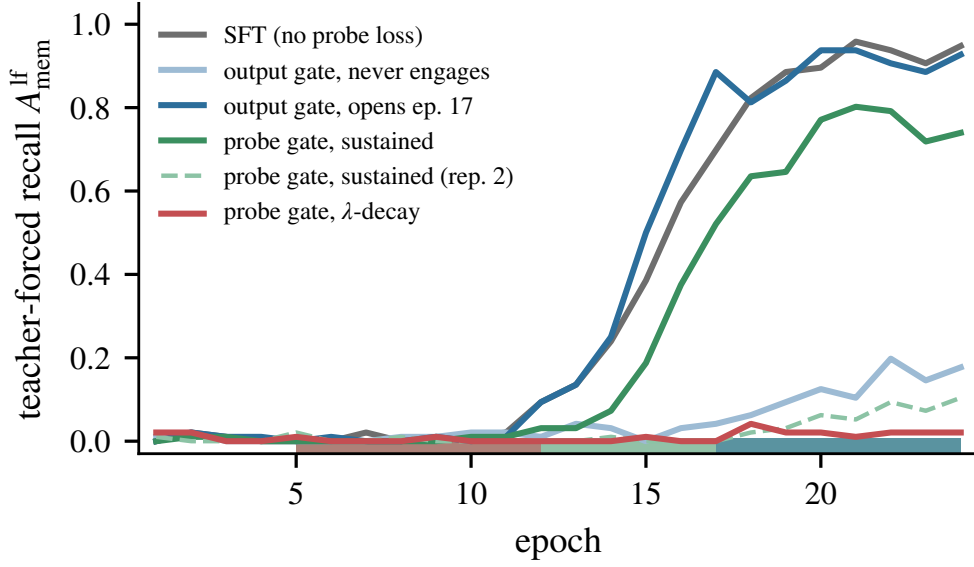


Figure 3: The completed engagement-schedule grid at 1.7B: final recall spans 0.012–0.934 under identical data, seed, and optimizer. Colored bars (bottom) mark each arm’s engagement window ($\lambda > 0$). SFT parity is achieved by late output-triggered engagement; early sustained or transient engagement does not reach it.

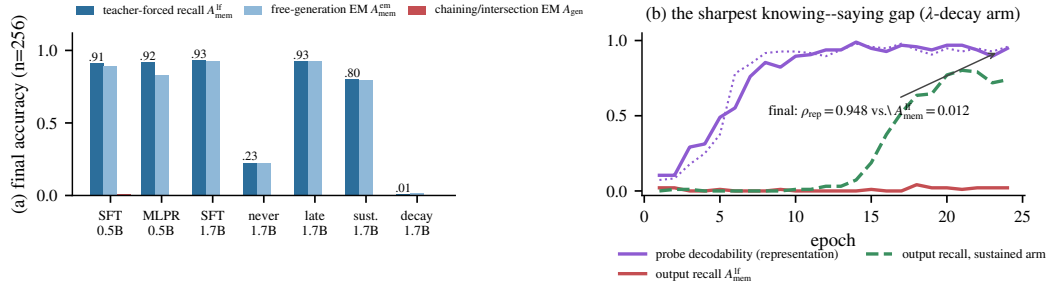


Figure 4: Knowing vs. saying across the grid. (a) Endpoints per run: teacher-forced recall (dark), free-generation EM (light), generalization EM (red). Every arm recalls; no arm generalizes. (b) The λ -decay arm: representation decodability reaches 0.948 while output recall stays at 0.012; the sustained arm shows the same representation trajectory paired with a normally rising output curve.

whose recall completes; the three arms still mid-transition at epoch 24 (λ -decay, sustained rep. 2, never-engaged) stall at 0.51–0.64 instead, and in the decay arm the stall sets in right after the probe weight is removed. Until their gates open, the MLPR arms’ CE is indistinguishable from their SFT controls: the detached probe is a pure observer during warm-up. The probe loss (Fig. 5b) falls from its random-head value (~ 17 –18) to 0.05–0.16, two orders of magnitude below chance $\ln 1600 \approx 7.38$, within 20 epochs at both scales.

Figure 5c overlays $\rho_{rep}(t)$ for all six MLPR arms. The representation saturates early everywhere: the probe crosses 0.5 between epochs 4 and 7 at both scales, while output recall crosses 0.5 only at epochs 15–17 in the arms that get there at all, and never in three arms (Fig. 5d). The probe’s head start is 13 epochs at 0.5B and 8–18 epochs at 1.7B—the measurable form of “knowing precedes saying”: the mid-layer representation organizes acquired facts long before the LM head reliably produces them.

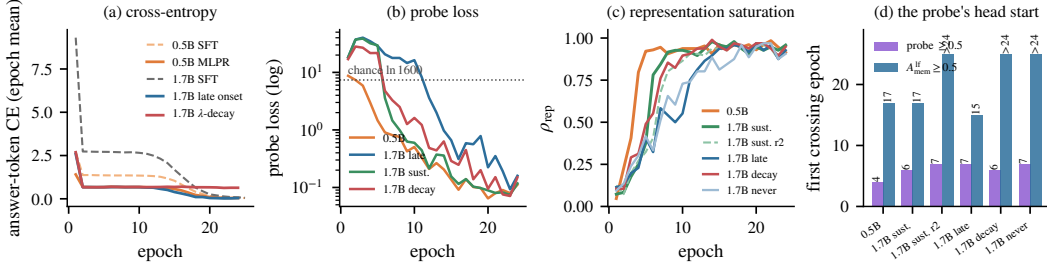


Figure 5: Dynamics and the early-warning probe. (a) Per-epoch mean CE: SFT controls (dashed) and MLPR arms (solid) overlap until the gate opens; the λ -decay arm’s CE stalls near its early plateau. (b) Probe loss (log scale) falls far below chance ($\ln 1600 \approx 7.38$) during detached warm-up. (c) $\rho_{\text{rep}}(t)$ for all MLPR arms: every curve saturates >0.9 regardless of the arm’s output outcome. (d) First crossing epoch of 0.5 for the probe (purple) vs. output recall (blue): the probe leads by 8–18 epochs, and leads arms whose output never crosses (>24).

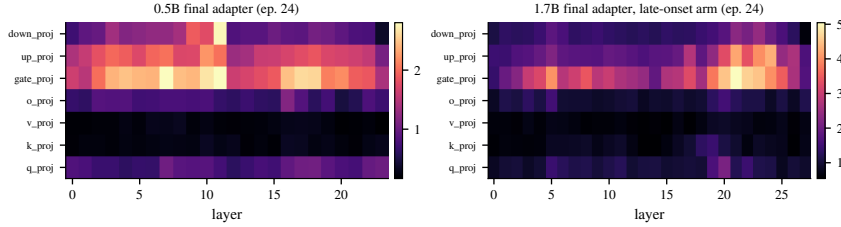


Figure 6: Update anatomy $\|\Delta W\|_F$ per (layer, module) from the final adapters. (a) 0.5B probe-gated arm; (b) 1.7B late-onset arm (gate open from epoch 17). MLP projections dominate at both scales; color scales are per-panel.

5.5 Update anatomy: upper-layer MLPs carry the memorization

From the released final adapters we compute $\Delta W = \frac{a}{r}BA$ per adapted projection (Fig. 6). MLP projections carry 71.1% (0.5B) and 65.0% (1.7B) of total update mass $\sum_l \|\Delta W\|_F$; key/value attention projections are nearly untouched; and the top three (layer, module) cells are gate/up-projection in every case—at 1.7B they are the top-layer gate projections (layers 21–23 of 28) of the post-activation arm, so the concentration holds after eight epochs of probe-gradient flow, not merely in an un-activated regime. At 0.5B the mass is more mid-layer (46.1% in the upper half vs. 55.1% at 1.7B).

6 Discussion

The knowing–using gap as a first-class object. Our results argue for treating the gap between internal recall and behavioral production as a measurable training-time quantity rather than a post-hoc evaluation surprise. The likelihood metric and the mid-layer probe are cheap (a forward pass per epoch; a linear head), and together they separate regimes that aggregate observables conflate: not-yet-known, known-but-unsayable, and—in the λ -decay arm—representation-known-but-output-never-formed.

The gate signal and its schedule are design axes. The same formal gate behaves very differently depending on what it watches and for how long. An output-likelihood gate defers engagement until the model can recall—principled, but its onset is stochastic (one replicate crossed at epoch 17; another never crossed), and a schedule that silently never fires is indistinguishable from plain SFT. The probe-saturation gate fires early at both scales because a fresh linear head measures representation organization rather than LM-head surface learning. What decided the outcome was not whether probe gradients flowed,

but when and for how long: late, output-triggered engagement at full weight matched SFT recall; early sustained injection was high-variance and below SFT; and transient injection that decayed to zero left the model at 0.012 recall while its mid layer organized the entities at 94.8% decodability—even though the arm then received twelve epochs of pure SFT and never recovered. Probe regularization should therefore engage after saturation and then persist; both halves of that prescription are testable with the released harness.

The saying gap is invariant to what we varied. No arm at either scale exceeded 1.2% free-generation EM or 0.4% generalization EM, while token-level recall under forcing reached 99%: the gap lives at the representation-to-surface interface, and closing it plausibly requires assembly-oriented objectives or decoding interventions rather than better-organized storage.

Controllers must not fight the phenomenon. The decay-spiral case study (Appendix D) is a small but concrete instance of a general design error: an adaptive controller aimed at a target regime treated that regime’s signature as a fault and drove the system away from it. The floor repair converts the controller from opposed to subordinate: damp instability, never extinguish the dynamics under study.

7 Limitations

Our findings are bounded in four ways. First, the corpus is synthetic and single-domain: crossing epochs and gap magnitudes will differ on natural knowledge (e.g., long-tail Wikipedia facts (Kandpal et al., 2023)). Second, each schedule is represented by one or two replicates, run-to-run variance at 1.7B is itself large (sustained arms: 0.109 vs. 0.805), and GPU non-determinism makes even same-seed runs diverge in takeoff epoch; our schedule claims rest on paired tests against a common SFT control, but single-run comparisons at this scale are unreliable. Third, the 24-epoch budget censors late-takeoff arms: the never-engaged replicate was still rising when it ended (0.227), so its endpoint is a lower bound. Fourth, the probe is linear and trained on the corpus it monitors, all experiments use LoRA at a single rank, and the mechanism behind the decay-arm collapse—like that of the sustained arms’ variance—is hypothesized rather than established; the released per-step telemetry is intended to make such follow-ups cheap.

8 Conclusion

We presented MLPR, a training-time instrumentation-and-control layer that makes the knowing—using gap measurable online and acts on it through a saturation gate. In a two-scale engagement-schedule grid we showed that the gate signal decides when engagement is possible (probe decodability fires at epochs 3–6, the output signal 11–16 epochs later or never), that the engagement schedule decides whether memorization completes at all (final recall 0.012–0.934 under identical data, seed, and optimizer, with late output-triggered engagement reaching SFT parity by McNemar test), captured the gate’s live re-evaluation transient, measured a knowing/saying dissociation that no schedule closes (generalization EM $\leq 0.4\%$ everywhere) and a representation-without-output regime induced by transient engagement, and confirmed from released adapters that memorization updates concentrate in upper-layer MLP projections. Transferring the instruments to natural corpora, scaling replicates beyond two per arm, and pairing the regularized phase with assembly-oriented objectives are immediate next steps.

Ethics Statement

This work studies memorization dynamics, which sit at the intersection of capability and privacy: the same teacher-forced recall metric that monitors learning could, in adversarial hands, audit models for memorized personal data, and the recall/production dissociation we document cautions that exact-match audits may underestimate what a model has stored. We release only synthetic entities invented for this study, and we believe better monitoring instruments ultimately serve privacy auditing and data governance more than they enable extraction.

Reproducibility Statement

All code (dataset generator, trainer, callbacks, gate and engagement schedules, controller, Modal deployment, analysis, McNemar module, and figure scripts) is included in the released codebase, together with the exact configurations of Appendix A. For every production run we release the final adapter and probe head, the per-epoch gate history, the complete stdout log, and a per-sample final evaluation record whose prompt-aligned correctness vectors are the direct inputs of every McNemar test reported here; per-epoch numbers in every figure and in Appendix C are traceable to these artifacts, identified by their Weights & Biases IDs in Appendix E. The earlier 50-epoch-class runs behind the decay-spiral case study and the scale-saturation pilots are the v4-generation artifacts also listed there.

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410 A Hyperparameters and Configurations

411 Table 2 lists the exact configuration shared by the production grid. All ten runs executed
 412 on Modal on a single NVIDIA A100 40 GB GPU in bf16, with the full stdout stream teed
 413 to a persistent volume; per-step and per-epoch curves, LoRA/probe matrix panels (every 4
 414 epochs), and run summaries are in the W&B project `knowing_using_gap`. Arm-specific gate
 415 settings are given in Table 1: probe-saturation gate ($\tau_0=0.35, \Delta_0=0.15$), output-likelihood
 416 gate ($\tau_0=0.8, \Delta_0=0.05$), decay horizon $D=8$ epochs, and $\lambda_0=0.3$ wherever the gate engages.

417 B Dataset Construction

418 The generator (released as `scripts/generate_synthetic_dataset.py`) proceeds in three steps.
 419 (1) Entity vocabulary: 1,600 invented entity names are generated from syllable templates;
 420 each is assigned an id in $[0, 1600)$ for the probe target space. (2) Memorization split $\mathcal{D}_{\text{mem}}$:
 421 1,200 question-answer pairs in a fixed interrogative template ending with Answer: label;
 422 the suffix is present during CE training (so the format is learned) and stripped during
 423 all evaluations (so it cannot leak the answer into the prompt). (3) Generalization split
 424 $\mathcal{D}_{\text{gen}}$ (450 prompts): 200 chaining prompts combine two trained facts into a compositional
 425 question, and 250 intersection prompts re-ask trained facts with paraphrased templates and
 426 surface forms never seen in training. Free-generation evaluations use left-padding, greedy
 427 decoding, and strict exact match after answer extraction; the likelihood evaluation (3) scores
 428 the first answer token and the token-level overlap under teacher forcing. The per-epoch
 429 gate evaluations run on a fixed subsample of 256 memorization prompts (32 batches) plus
 430 the generalization prompts, keeping epoch overhead at ~ 1 –2 minutes; the final evaluation

Parameter	0.5B runs	1.7B runs
Model	Qwen2.5-0.5B-Instruct	Qwen3-1.7B
LoRA r / α / dropout	16 / 32 / 0.05	16 / 32 / 0.05
LoRA targets	q,k,v,o,gate,up,down	q,k,v,o,gate,up,down
Probe layer l^*	12 of 24	14 of 28
Probe head	896 $\rightarrow$ 1600	2048 $\rightarrow$ 1600
Trainable params	8.8M (1.75%)	17.4M (1.00%)
Backbone LR	2×10^{-4}	2×10^{-4}
Probe LR	2×10^{-3}	2×10^{-3}
Schedule	cosine, warmup 5%	cosine, warmup 5%
Batch (dev $\times$ accum)	8×2	4×4
Steps per epoch	75	75
Epochs	24	24 (two 16-ep. censored)
Seed	42	42
λ_0	0.3	0.3
Gate signals	$\rho_{\text{rep}}, A_{\text{mem}}^{\text{lf}}$	$\rho_{\text{rep}}, A_{\text{mem}}^{\text{lf}}$
Decay horizon D	—	8 epochs (λ -decay arm)
Precision / GPU	bf16 / A100-40GB	bf16 / A100-40GB
Matrices to W&B	every 4 ep	every 4 ep

Table 2: Full hyperparameter configurations for the production grid.

of every run scores $n=256$ prompts per channel (memorization generation, memorization teacher-forced, and 174 chaining + 82 intersection generalization prompts) and dumps the per-sample records used by the McNemar tests.

C Full Per-Epoch Curves

Table 3 reports the 0.5B pair and Table 4 the 1.7B grid every 4 epochs (the complete per-epoch JSON histories are in the released telemetry, one gate_history.json per run). Recall values are the per-epoch gate evaluations on the 256-prompt subsample; the final-evaluation endpoints in Table 1 are the independent $n=256$ evaluations at run end, which agree with the last gate epoch to within ± 0.05 . The two censored 16-epoch runs (SFT and MLPR at 1.7B) reproduce the early-time trajectories of their 24-epoch counterparts and end at final evaluations of $A_{\text{mem}}^{\text{lf}}=0.320$ (SFT, still rising) and 0.035 (MLPR, probe gate open from epoch 4).

ep	SFT $A_{\text{mem}}^{\text{lf}}$	MLPR $A_{\text{mem}}^{\text{lf}}$	λ	ρ_{rep}
1	0.000	0.000	0.000	0.047
4	0.008	0.000	0.300	0.797
8	0.016	0.000	0.300	0.906
12	0.016	0.008	0.300	0.938
16	0.469	0.352	0.300	0.930
20	0.945	0.844	0.300	0.930
23	0.938	0.906	0.300	0.945
24	0.922	0.891	0.300	0.914

Table 3: 0.5B pair (probe-gated MLPR vs. SFT). The probe signal crosses $\tau_0=0.35$ at epoch 3 and the gate holds $\lambda_0=0.3$ from epoch 4; output recall takes off around epoch 14–16 in both arms and the endpoints are statistically indistinguishable (McNemar $p=0.84$).

D The Decay-Spiral Failure and Its Repair

Failure. In a pilot 50-epoch 0.5B run (v4 generation), the eval-aware controller issued eval-driven decays every two epochs for five consecutive episodes while held-out loss diverged, driving the cumulative LR multiplier along $1.0 \rightarrow 0.49 \rightarrow 0.34 \rightarrow 0.24 \rightarrow 0.17$, breaching the controller’s own nominal minimum (0.3) because the eval-driven pathway applied decays without consulting the instability path’s floor. Memorization was choked: likelihood recall

ep	SFT	never	late	sust.	decay	sust. 2
1	0.000	0.010	0.000	0.000	0.021	0.010
4	0.000	0.000	0.010	0.000	0.000	0.000
8	0.000	0.010	0.000	0.000	0.000	0.010
12	0.094	0.010	0.094	0.031	0.000	0.000
16	0.573	0.031	0.698	0.375	0.000	0.000
20	0.896	0.125	0.938	0.771	0.021	0.062
24	0.948	0.177	0.927	0.740	0.021	0.104
$\lambda(t)$ for the MLPR arms:						
8	—	0.00	0.00	0.30	0.19	0.30
12	—	0.00	0.00	0.30	0.04	0.30
16	—	0.00	0.00	0.30	0.00	0.30
20	—	0.00	0.30	0.30	0.00	0.30
24	—	0.00	0.30	0.30	0.00	0.30

Table 4: 1.7B engagement-schedule grid ($A_{\text{mem}}^{\text{lf}}$ per epoch; λ below). “never”: output gate never crossed (max 0.198); “late”: output gate opened at epoch 17; “sust.”: probe gate from epoch 5 (rep. 2 from epoch 6); “decay”: probe gate with linear decay to zero over $D=8$ epochs from onset.

449 stayed pinned near 0.02 and the gate threshold became unreachable. The run was stopped
450 to avoid burning compute on dynamics that the controller itself had frozen.

451 Diagnosis. The failure is structural, not incidental: in the memorization regime, held-out
452 divergence is the expected signature that the saturation gate waits for, so an eval-driven
453 damping rule without a floor is guaranteed to eventually fight the gate.

454 Repair. The eval-driven pathway now clamps the cumulative multiplier to $m_{\text{min}}=0.5$ (Eq. 6)
455 and refuses further eval-driven decays at the floor. Validation proceeded in three stages: (i)
456 unit tests on a stubbed trainer assert the floor is respected exactly (m lands on 0.5, never
457 below; optimizer LRs match base $\times$ floor); (ii) a dedicated 5-epoch GPU run with an
458 unreachable gate reproduced the failing regime ($\lambda=0$ throughout, eval rising) and showed
459 one eval-driven decay ($1.0 \rightarrow 0.7$) with the floor never breached; (iii) the 50-epoch-class
460 production runs that used the controller clamped at 0.5 and held for the remainder of
461 training while memorization proceeded unimpeded. The present grid fixes the LR schedule
462 (cosine) so that the engagement schedule is the only active intervention; the controller ships
463 with the stack and its decisions are logged whenever it is enabled.

464 E Artifact Map

465 Production grid (this paper). Each of the ten runs is identified by a W&B run in project
466 knowing_using_gap (entity codewithdark-islamia-college-peshawar) and a run directory
467 on the persistent volume holding: summary.json (endpoint block), gate_history.json (per-
468 epoch gate evaluations), eval_details.json (256 per-sample final records per channel with
469 em/contains/token_f1/ token_acc and eval_type, the McNemar inputs), training_log.txt
470 (complete stdout with per-step loss dicts), adapter_model.safetensors, and probe_head.pt.
471 Run IDs: 8nqeb64s (0.5B SFT), 8bbmxqsy (0.5B MLPR), t363oowy (1.7B SFT censored),
472 9s50ruuf (1.7B SFT), 88gpa338 (1.7B MLPR censored), kxjrtb71 (1.7B probe-gate rep. 2),
473 ugii4c9x (1.7B output-gate never-crossed), obcthrqo (1.7B λ -decay), myrqofvo (1.7B probe-
474 gate rep. 1), vx5w4obp (1.7B output-gate late-onset). Earlier-generation artifacts (v4). The
475 50-epoch 0.5B run and the budget-truncated large-model run behind the decay-spiral case
476 study (Appendix D) and the scale-saturation pilots cited in Section 5.1 are on the Hugging
477 Face Hub (user codewithdark) as mlpr-qwen2.5-0.5b-instruct-50ep-adaptive-v4 and mlpr-
478 qwen2.5-7b-instruct-50ep-adaptive-v4, with gate and controller histories and full training
479 logs. Codebase: the released zip contains the full training stack (src/ with probe, LoRA
480 setup, gate callback, engagement schedules, adaptive controller, likelihood evaluator, W&B
481 logger), the Modal deployment (modal_train.py), the dataset generator, the analysis and
482 figure scripts (including the McNemar module), and this paper under paper/.

483 F The Likelihood Recall Metric in Detail

484 For a memorization pair (x, y) the prompt is built by stripping the Answer: suffix from
485 the training text. The metric performs a single teacher-forced forward pass over prompt $\oplus$
486 answer with left padding and records (i) strict first-answer-token exact match (3), and (ii) the
487 fraction of answer tokens whose greedy next-token prediction matches the reference (token
488 recall; token_acc in the released records). Both are macro-averaged over the subsample.
489 Compared with free-generation EM, this removes decoding length, format, and repetition
490 effects. In the production grid the strict teacher-forced EM and substring containment
491 coincide on every arm (the free generations either emit the exact entity string or miss
492 entirely), so the storage-vs-assembly contrast is carried by token recall (0.99 in saturated
493 arms) versus free-generation EM (≤ 0.012 everywhere): the identity is stored and linearly
494 decodable, and its first answer token is one greedy step away, yet free decoding almost never
495 assembles it.